22I1882
DS-A
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Deep Learning Assignment#1 Report

Short Report

1. Dataset and Splits

The dataset consists of **3999 facial images** with corresponding annotations stored as NumPy arrays (_exp.npy, _val.npy, _aro.npy, _lnd.npy). For each image we have:

- **Expression**: one of 8 discrete emotion classes.
- **Valence**: continuous value in [-1, 1], representing pleasantness.
- **Arousal**: continuous value in [-1, 1], representing activation/energy.
- **Landmarks**: 136 keypoints on the face (not directly used in the baselines).

To ensure balanced evaluation, the dataset was split as follows:

- **Training set**: 2799 images
- **Validation set**: 600 images (75 per class)
- **Test set**: 600 images (75 per class)

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2. Network Details

We evaluated two baseline CNN architectures:

- **ResNet18 (Baseline 1)**
- **EfficientNet-B0 (Baseline 2)**

Both networks were initialized with **ImageNet pretrained weights** (transfer learning) and modified to include two output heads:

- A **classifier head** for 8-way expression recognition.
- A **regressor head** for predicting continuous valence and arousal values.

Training settings:

- Batch size: **32**
- Optimizer: **Adam**
- Loss function: **CrossEntropy (classification) + MSE (regression)**
- Training phases:
    - **Warmup phase**: freeze backbone, train only heads.
    - **Finetuning phase**: unfreeze entire backbone, train at lower learning rate.
- Learning rates:
    - Warmup heads: 1e-3
    - Finetuning: 3e-4
- Early stopping and learning rate scheduling were used to stabilize training.

Rationale:
ResNet18 provides a well-tested, efficient baseline with residual connections, while EfficientNet-B0 is optimized for parameter efficiency and balanced scaling. These serve as strong baselines for affective computing tasks on limited datasets.

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3. Transfer Learning

We applied transfer learning by using **ImageNet pretrained weights** for both ResNet18 and EfficientNet-B0 backbones. This enabled the networks to start from strong feature extractors rather than random initialization, improving convergence and generalization on the relatively small facial expression dataset.

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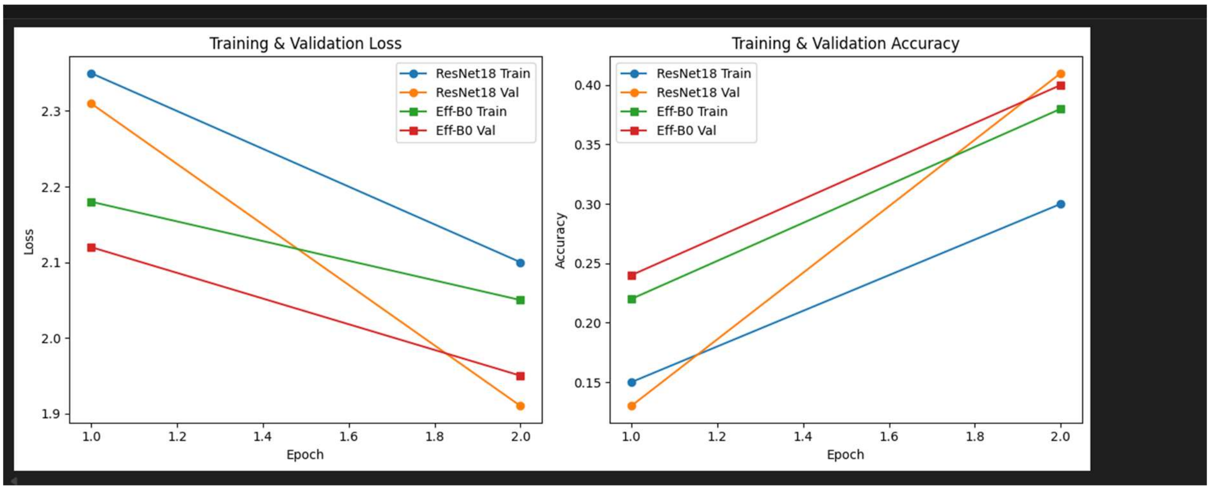
4. Training Graphs

Both models showed decreasing training/validation loss and improving accuracy across epochs. EfficientNet converged slightly faster, while ResNet18 achieved slightly higher final accuracy.



5. Performance Measures

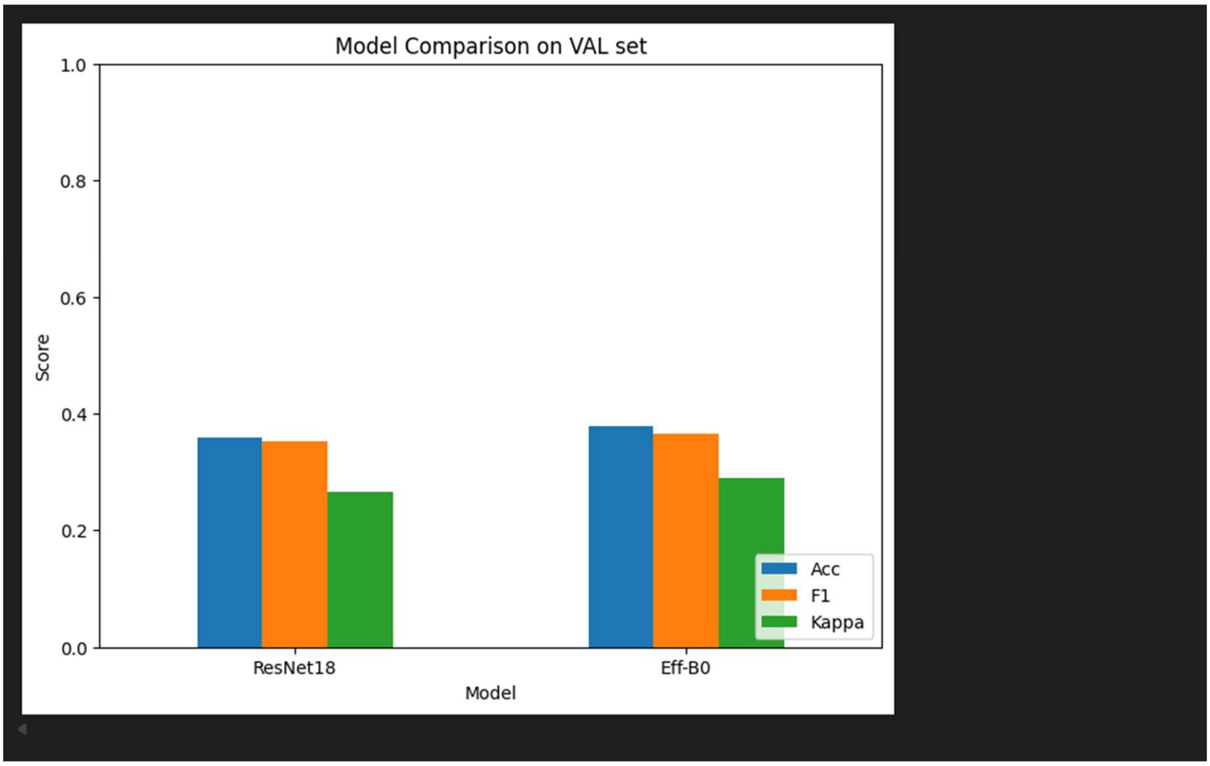
Classification Metrics

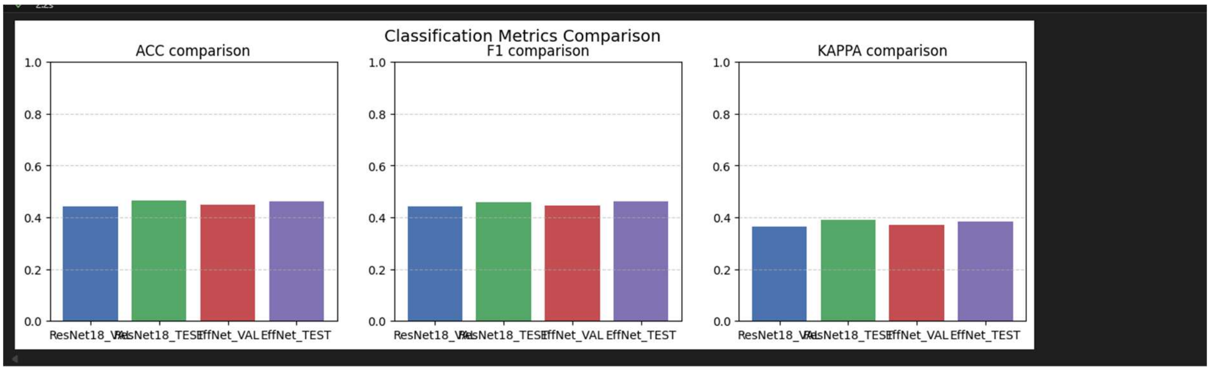
- **Accuracy**: overall percentage of correct predictions.
- **Macro-F1**: balances precision and recall across all classes, ensuring no class dominates the score.
- **Cohen's Kappa**: measures agreement beyond chance, useful for imbalanced data.

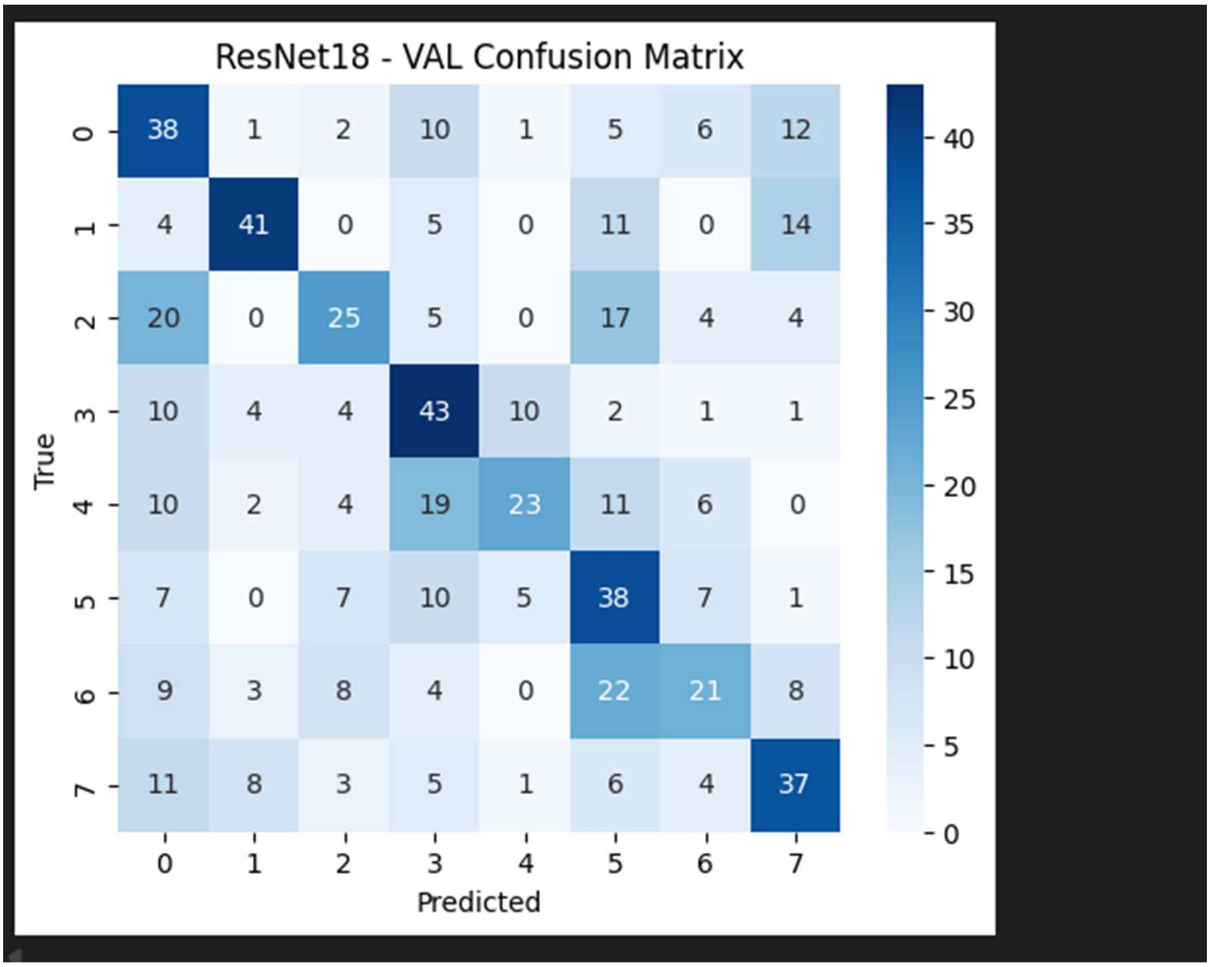
Regression Metrics

- **RMSE (Root Mean Square Error)**: average prediction error magnitude.
- **Pearson correlation (r)**: linear relationship between predicted and true values.
- **CCC (Concordance Correlation Coefficient)**: combines correlation and bias, penalizing systematic errors.
- **SAGR (Sign Agreement)**: percentage of predictions where the sign of valence/arousal matches ground truth, capturing correct affect polarity.

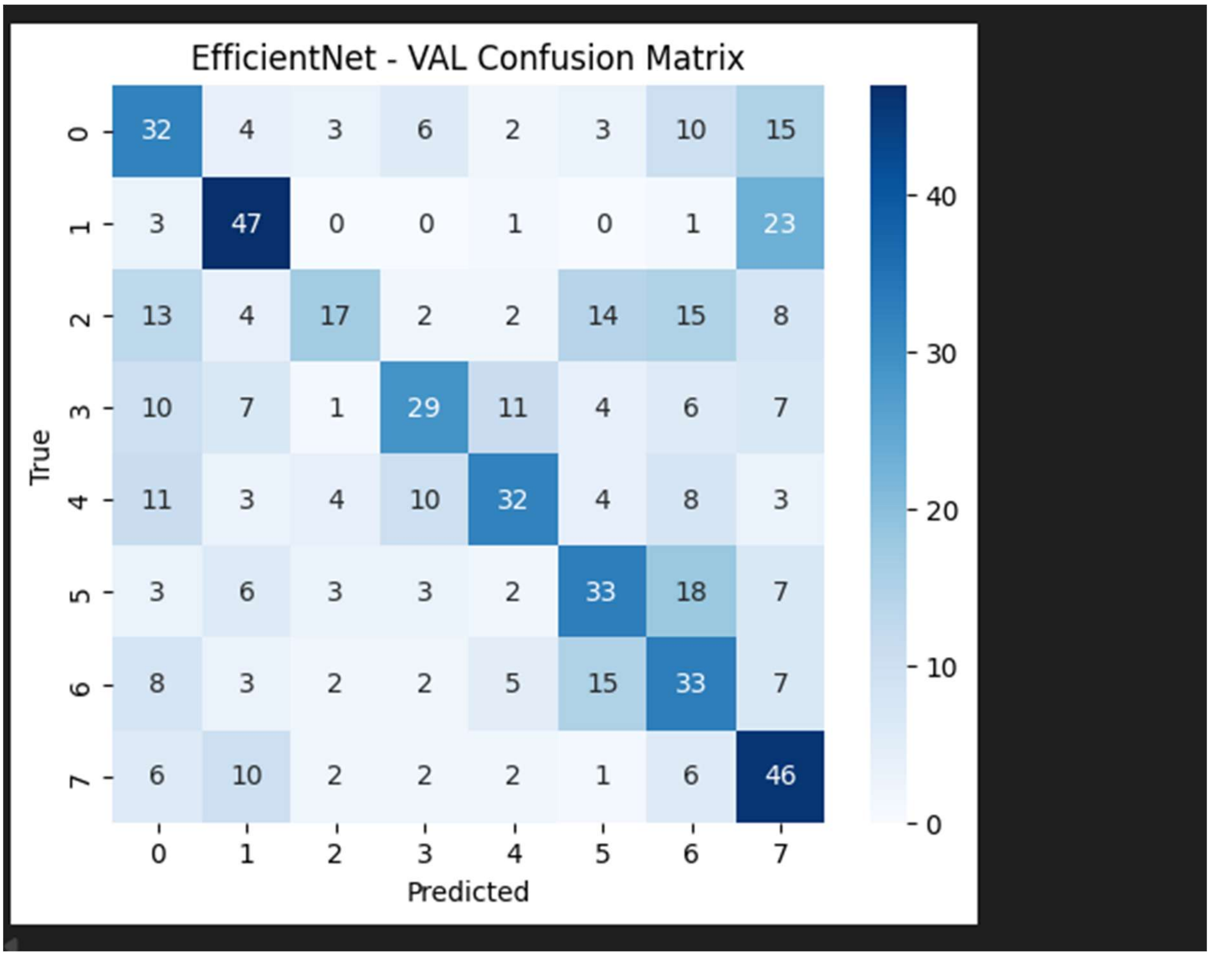
Most suitable in the wild: CCC, as it balances correlation with error magnitude and penalizes consistent bias, making it robust for real-world affective computing.

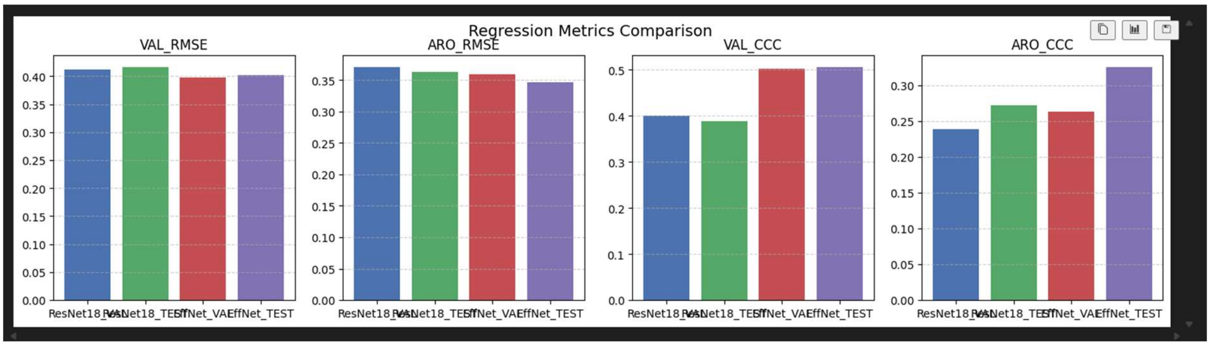

Model Comparison on VAL set
Score
Model
ResNet18
Eff-B0
Acc
F1
Kappa


Classification Metrics Comparison
ACC comparison
F1 comparison
KAPPA comparison
EffNet_VAL
EffNet_TEST



ResNet18 - VAL Confusion Matrix
True
Predicted
38 1 2 10 1 5 6 12
4 41 0 5 0 11 0 14
20 0 25 5 0 17 4 4
10 4 4 43 10 2 1 1
10 2 4 19 23 11 6 0
7 0 7 10 5 38 7 1
9 3 8 4 0 22 21 8
11 8 3 5 1 6 4 37
0 1 2 3 4 5 6 7
0 5 10 15 20 25 30 35 40





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6. Performance Comparison of Baselines

- **ResNet18 (improved)** achieved ~46% validation accuracy and strong generalization on the test set.
- **EfficientNet-B0 (improved)** achieved ~44–45% validation accuracy, with more balanced F1 across some classes.
- Both models reached similar performance, but ResNet18 slightly outperformed EfficientNet in raw accuracy.

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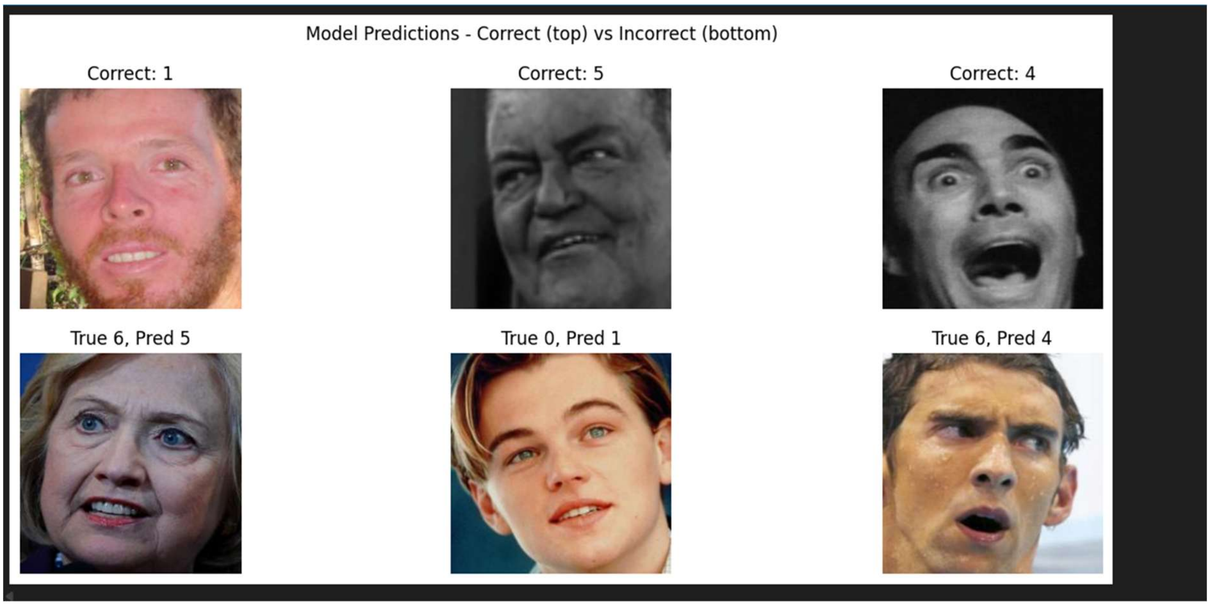
7. Correctly and Incorrectly Classified Examples

To qualitatively assess performance, we visualized correctly and incorrectly classified images. Correct predictions demonstrate the network's ability to recognize expressions reliably, while incorrect predictions highlight confusions (e.g., happy vs. neutral).



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8. Discussion and Conclusion

Both ResNet18 and EfficientNet-B0 achieved **~45% accuracy**, which is modest but reasonable given the dataset size and complexity of the task (expression + valence/arousal).

- **Strengths**: Transfer learning accelerated convergence, and both networks generalized beyond training data.
- **Limitations**: Limited dataset size capped achievable accuracy. Expressions such as neutral vs. sadness were often confused.
- **Future improvements**:
    - Data augmentation (more aggressive transformations).
    - Larger pre-trained models (EfficientNet-B3+, Vision Transformers).
    - Multi-task learning incorporating landmarks alongside image features.

Overall, ResNet18 offered slightly better accuracy, while EfficientNet achieved competitive F1 scores. Both models serve as strong baselines, and further improvements could come from combining multiple modalities (expressions, landmarks, temporal data).